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Learning Diary
R504TL194 Cloud Computing

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6 February 2026

R8 Student's written exercises

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R10 Module 1 assignment. Online learning introduction and teachable machine in use

Module 1 lectures introduce the fundamental concepts of cloud computing, cloud service providers, learning tools, and the basics of machine learning in cloud environments. The Module 1 assignment consists of two parts: completing an online cloud learning module (Amazon AWS, Google Cloud, or Microsoft Azure) and creating a basic machine learning (ML) model using Teachable Machine. I chose Microsoft Azure because it is frequently listed as a skill in IT job postings.

Lecture synthesis

The cloud is a vast online storage space where people and businesses store their files and applications, accessible from anywhere with an internet connection. The main purpose of the cloud is to provide on-demand access to computing resources and services over the internet. This includes a wide range of services such as servers, storage, databases, networking, software, analytics, and intelligence (Microsoft Azure 2026). Cloud computing is the delivery of computing services – including servers, storage, databases, networking, software, analytics, and intelligence – over the internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale (Microsoft Azure 2026).

The topic of cloud computing services is Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) (see Figure 1) (Chen 2020). To understand the differences between various cloud service models, look at Figure 2 (Tripathy 2021).

Top Cloud Service Providers in 2026 is Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM Cloud and Oracle Cloud Infrastructure (OCI) (Martin 2025).

(Mikko Pajula) According to the presentation, while Cloud cost management remains complex and limits pedagogical choices, provider-specific estimation tools are essential for effective resource planning (Pajula 2026). Then the lecture provided an overview of instructional resources and estimation tools (Pajula 2026).

The lecture illustrated that ML is a subset of Artificial Intelligence (AI), with Neural Networks and Deep Learning (DL) forming a nested hierarchy within it, as shown in the Figure 3 (Pajula 2026). The ML process is a systematic pipeline involving data acquisition, complex preprocessing, and computationally intensive training, followed by model evaluation and deployment in production environments (Pajula 2026).

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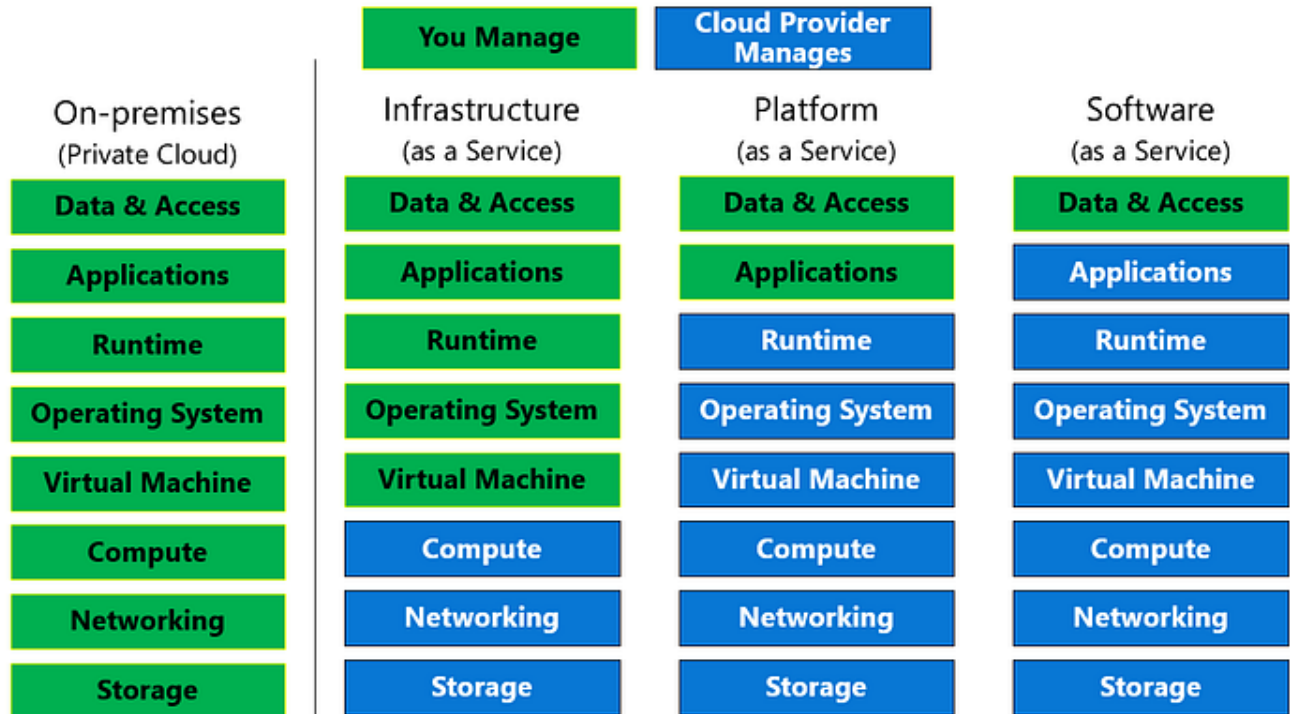


Figure 1. The three major service categories provided by cloud providers: what they are and what is the difference between them (Chen 2020)

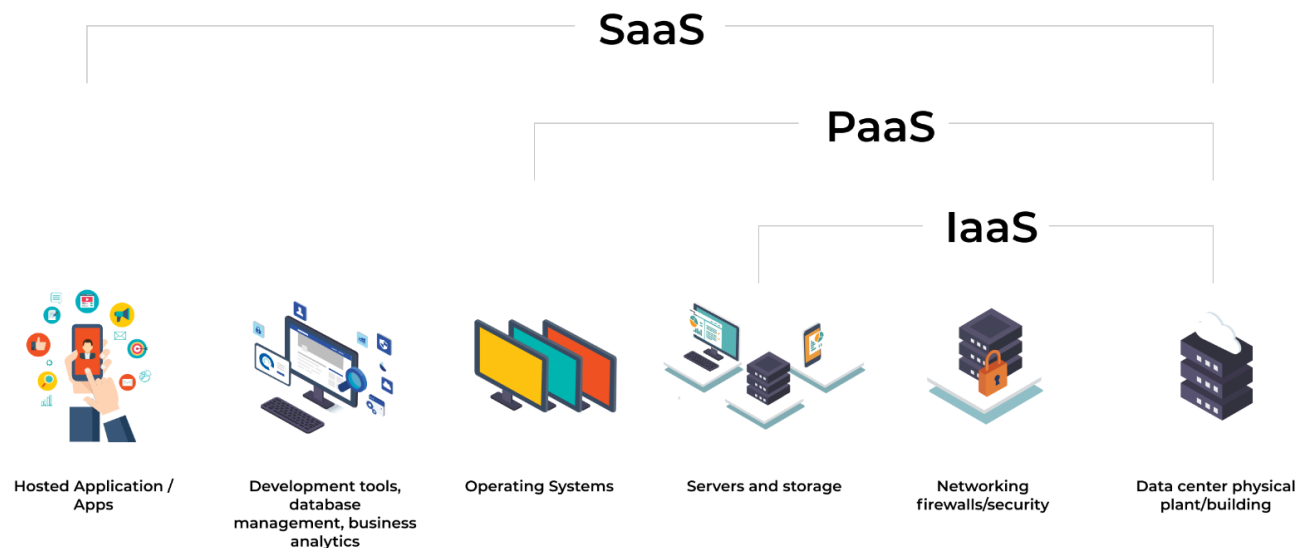


Figure 2. IaaS, PaaS and SaaS each manage different elements of cloud computing (Tripathy 2021)

I learned that while the cloud solves on-premise hurdles like high costs and limited scalability, it introduces its own complexities such as the "memory wall" and usage-based pricing. AutoML tools, like Google Teachable Machine, make the machine learning process more accessible by automating difficult manual steps. Platforms like

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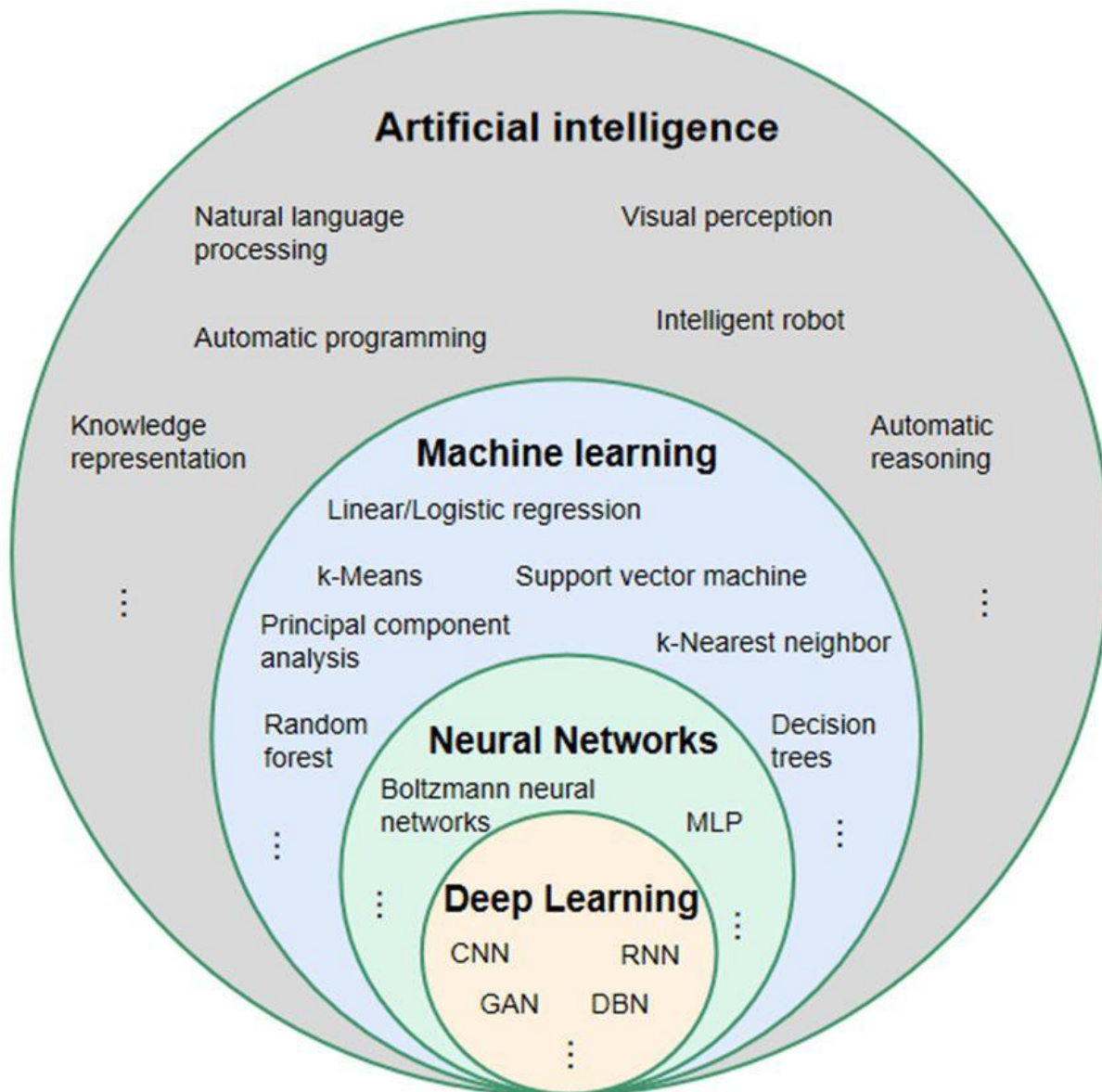


Figure 3. The difference between AI, ML and DL (Vyacheslav 2023)

Google Colab provide an essential entry point for students by offering free, cloud-based access to high-performance computing resources (GPUs/TPUs) without the need for complex local setup.

Microsoft Azure: Introduction to generative AI and agents

The module I selected was designed for 37 minutes of study (7 Units) – [link](#).

Generative AI operates through Large Language Models (LLMs) that process natural language by breaking it down into tokens and converting them into numerical vectors called embeddings. These embeddings capture semantic meanings, allowing the model to understand relationships between words. The core architecture, the Trans-

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former, utilizes an attention mechanism to weigh the importance of different tokens in a sequence, enabling the model to predict the most probable and contextually relevant completion for any given prompt.

Effective interaction with these models relies on prompt engineering, where system prompts define the model's behavior and user prompts provide specific instructions. Advanced applications use Retrieval Augmented Generation (RAG) to ground responses in external data, ensuring accuracy. Beyond simple chat, the technology evolves into AI agents – autonomous systems that combine LLMs with specific instructions and tools to perform tasks and collaborate in multi-agent environments. Practical exercises demonstrate that adjusting model parameters like "temperature" and providing structured context are essential for optimizing AI output (Microsoft. 2026).

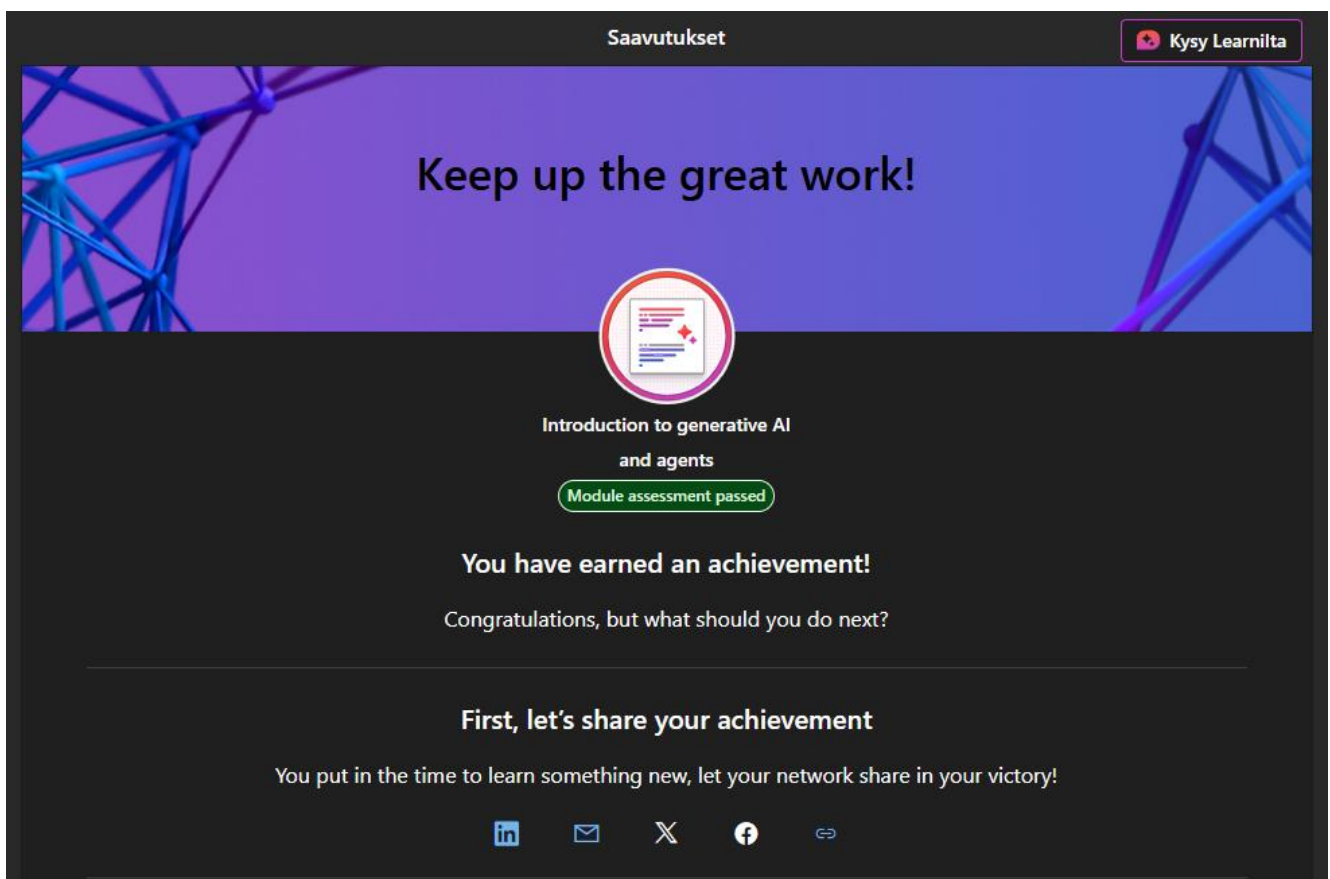


Figure 4. Introduction to generative AI and agents

Basic ML model using Teachable Machine

I chose the image project to complete the second part of Module 1 assignment. Created 6 classes: Glass, Cup, Table, Chair, Sofa, and Candle. I downloaded a set of images from the Internet and uploaded them for each class. I ran "Train Model" and then "Export Model",

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selecting the "TensorFlow.js" tab and "Download my model", downloading the "tm-my-image-model.zip" file containing "metadata.json", "model.json" and "weights.bin". Next, I used AI Gemini to generate HTML-code for my task: the model needs to recognize whether a frame contains a Glass, Cup, Table, Chair, Sofa, or Candle. The page displays a list of furniture items; as soon as the corresponding item appears on camera, it turns green. I created an empty file "index.html", placed the code recommended by AI Gemini into it, created a new folder in the same folder as "index.html" called "my_model" and moved the contents of "tm-my-image-model.zip" into this folder. To run the code, I opened index.html using Visual Studio Code, which already had Live Server installed. Right-clicking on index.html, I selected "Open with Live Server" from the context menu. The result can be seen in Figure 5.

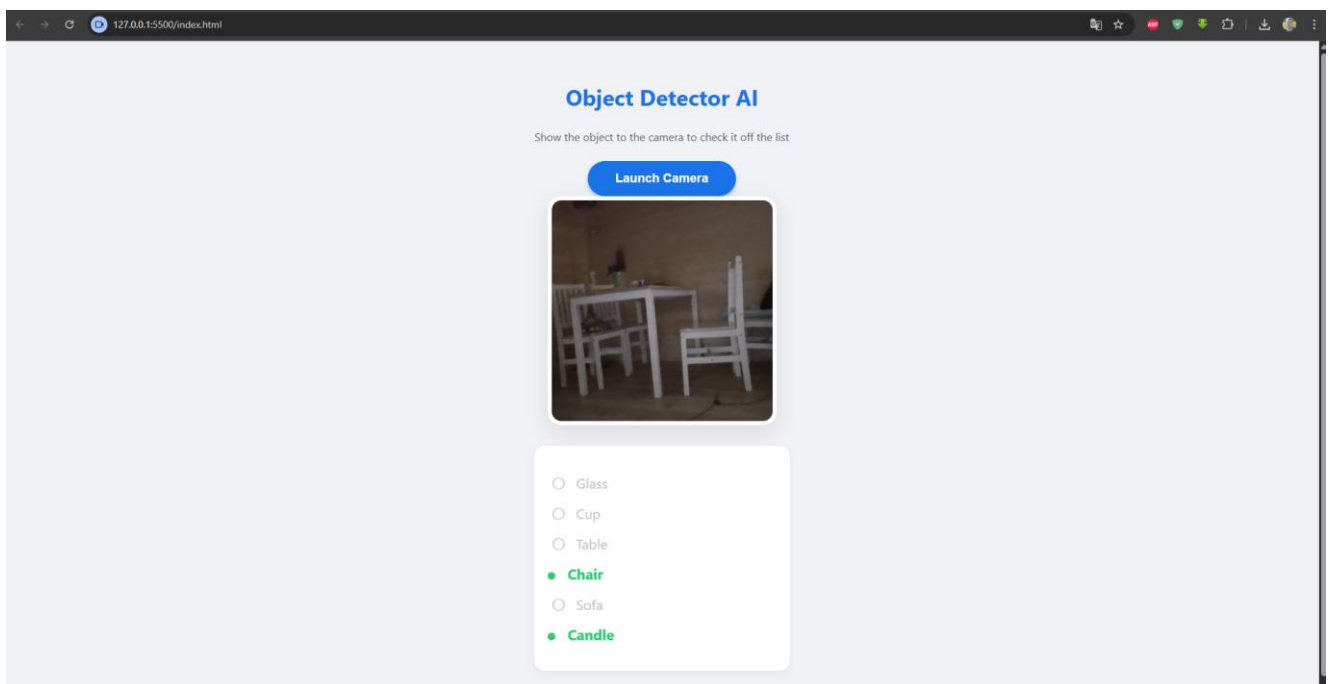


Figure 5. The result of running the HTML code

On the table, the model recognized the candle, and when I moved away, it recognized the chair, but no matter how hard I tried, the model did not recognize the table. Most likely, the failure is due to the fact that there are many things placed on the table, since the teaching materials used include many similar table models.

Conclusions

This learning diary summarizes the completion of the Module 1 assignment of the Cloud Computing course. The module provided a clear introduction to cloud computing fundamentals, including cloud service models, major cloud providers, and the basics of machine learning in cloud environments.

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Completing the Microsoft Azure online learning module improved my understanding of cloud-based AI services and their practical use. The hands-on task with Google Teachable Machine helped me understand the basic machine learning workflow, from data collection and model training to deployment in a simple HTML application.

Overall, Module 1 successfully connected theory with practice and provided a solid foundation for further studies in cloud computing and machine learning.

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6.2.2026 https://www.linkedin.com/posts/eric-vyacheslav-156273169_the-difference-between-artificial-intelligence-activity-7014145513159569408-nPBo/.

A handwritten signature in black ink, appearing to read 'Maksym Kotlubaiev'.

Maksym Kotlubaiev